

Granthōmukhīm
𑌕𑍇𑌃𑍆𑌒𑍊𑌖𑍅𑌔𑍄

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Chapter 1

The Language

Welcome to this treatise on the Granthōmukhīm language! This is my first conlang which has been in the works for about six years now, with hiatuses in the middle. It started out as an attempt to iron out the irregularities in Pāṇinian Sanskrit (ironic, isn't it?). I took the base vocabulary of Sanskrit and studied the Indo-Aryan and Indo-European branch of languages for information on how all this worked. The very early versions were just Sanskrit words with modified affixes and inflections. It was still Sanskrit, just more Ithkuil-like (I did fall prey to that fallacy). It took a few more years of refinement until the grammar matured enough, so I could wean it off of the ad-hoc Sanskrit lexicon base and introduce a native system. This was also around the same time that I was learning more about the Dravidian languages, as a tangent to my school language courses. The influence of Dravidian is clearly seen in the phonology that developed over time in this language, giving rise to a rich collection of short vowels, retroflexes, liquids and approximants.

While this was in progress, I made an attempt at a highly analytic language with no inflections or fusional grammar. The entire lexicon was agonistic to the grammar and meaning was achieved by means of particles, prepositions and postpositions. Its vocabulary was not exactly well-researched but more of me rambling and trying to emulate Greek phonology. I did end up using the Greek alphabet for the language. It eventually turned out futile and died out, but what it did teach me was the ease of making an inflectional language with free word-order.

Granthōmukhīm is an Indo-Dravidian language with its lexicon from Indo-Aryan origins, and a highly Dravidian phonology and phonotactics. It has a root-affixing system wherein atomic base words (called *roots*) are affixed to form *stems* of various classes, which are further injected with nuance and meaning, and finally grammatically inflected before being dropped in a sentence.

Here are the features and characteristics of this Granthōmukhīm:

- Phonemic length, aspiration and vowel quality provide semantic distinction.
- 10 grammatical cases: Nominative, Accusative, Instrumental, Ablative, Dative, Genitive, Inessive, Adessive, Subessive, Vocative
- 3 grammatical genders (which need not be biological and are purely semantic): Masculine, Feminine, Neuter
- 3 grammatical numbers: Singular, Dual, Plural
- A base-10 number system written successively from highest to lowest place, with the units preceding the next lowest place
- Highly agglutinative with morphosyntactic alignment between adjectives, verbs, nouns and pronouns.
- 3 tenses: Present, Past, Future
- 6 aspects: Gnostic, Habitual, Progressive, Aorist, Retrospective, Simple
- 7 moods: Declarative, Conditional, Imperative, Abilitative, Exclamatory, Optative
- Active and passive voices
- Nominal-derived case-based interrogatives and relatives
- 4-way distinction in proximity and in clusivity
- Internal and external sandhi

Chapter 2

Writing System & Phonetics

2.1 The Extended Brahmi Abugida

On paper, the language has a native Brahmic script, derived from inspecting and taking inspiration from all the modern Brahmic scripts. It is highly symmetric, with aspiration written as a secondary sound change and possesses regular consonant clusters in writing. While this is to be written down by hand, there's no Unicode support for a conscript, and hence I needed another script to write it in digital media. Incorporating a digital writing system to complement the spoken standard was a major challenge. In fact, I used Devanagari in the very early stages of the language's development. But I decided that standard Devanagari was lacking and something more substantial was needed to support the language, given its tendency to form long agglutinative words. Given its Indo-Aryan origins, the script ideally should have been a Brahmic abugida. But, for proof of concept, I used an extended version of the modern Hebrew alphabet, repurposing its letters to support the entire phonetic inventory of my language, including vowels, which is not exactly easy, given that Hebrew is an abjad. I made it a partial abugida by requiring every consonant to have an inherent vowel, but wrote down every other vowel beside by manual combination of the aleph and the yod sounds. The geresh was repurposed to cancel the vowel marker. It worked well, although reading it was not exactly easy, for I had no 'training' in reading Hebrew. Eventually, I used the Brahmi alphabet itself, but with a few extensions from Old Tamil. Despite being Indo-Aryan in origin, the phonology of the language is highly influenced by Dravidian, and thus possesses a rich set of liquids and retroflexes, and a lot of short vowels, which the standard Brahmi script does not account for.

2.1.1 Vowels

𑌕	x	Δ
𑌖	x	𑌗
𑌘	2	𑌙
𑌚	2	𑌛
𑌜	𑌝	𑌞
𑌟	Δ	

2.1.2 Consonants

+	d	c	λ	l	𑌐	𑌑	𑌒
𑌓	𑌔	𑌕	𑌖	𑌗	𑌘	𑌙	
𑌛	𑌜	𑌝	𑌞	𑌟	𑌠	𑌡	
𑌣	𑌤	𑌥	𑌦	𑌧	𑌨	𑌩	
𑌫	𑌬	𑌭	𑌮	𑌯	𑌰	𑌱	

2.1.3 Combinations

+	+	+
+	+	+
+	+	+
+	+	+
+	+	+
+	+	+

2.1.4 IPA Classification

	voiceless			voiced									
<i>glottal</i>					ɦ					a	a:		
<i>velar</i>		k	k ^h	g	ɡ ^ɦ	ŋ							
<i>palatal</i>	ɟ	Ɑ	Ɑ ^h	ɗ	ɗ ^ɦ	ɲ	j			i, e	i:	e:	ai
<i>retroflex</i>	ʂ	ɬ	ɬ ^h	ɖ	ɖ ^ɦ	ɳ		ɻ	ɭ	ɹ	ɹ:		
<i>dental</i>	s	ɸ	ɸ ^h	ɖ	ɖ ^ɦ	n		ɾ, r	l	ɭ	ɭ:		
<i>labial</i>		p	p ^h	b	b ^h	m	v			u, o	u:	o:	au
	<i>fric.</i>	<i>un.</i>	<i>asp.</i>	<i>un.</i>	<i>asp.</i>	<i>nas.</i>	<i>sv.</i>	<i>rh.</i>	<i>lat.</i>	<i>sh.</i>	<i>long.</i>		
		<i>plosive</i>					<i>gl.</i>	<i>liquid</i>		<i>mono.</i>		<i>diph.</i>	
							<i>approximant</i>			<i>vocalic</i>			
	<i>consonant</i>									<i>vowel</i>			

	voiceless			voiced									
<i>glottal</i>					ɬ					ɦ	ᶑ		
<i>velar</i>		ɸ	ɣ	ʌ	ɮ	ɮ̪							
<i>palatal</i>	ʌ	ɖ	ɖ̪	ɛ	ɸ	ᶑ	ɹ			ɹ̥, ɹ̥̪	ɹ̥̪	Δ	Δ̪
<i>retroflex</i>	ɬ	ɮ	ɮ̪	ɹ̥	ɮ̪	ɮ̪		ɹ̥̪	ᶑ̪	x	x̪		
<i>dental</i>	ɬ̪	ʌ̪	ɔ̪	ɹ̥̪	ɮ̪	ɮ̪		ɹ̥̪, ɹ̥̪̪	ᶑ̪	ʒ̪	ʒ̪̪		
<i>labial</i>		ɬ	ɮ	ɔ̪	ɮ̪	ɮ̪	ɔ̪			ɮ̪, ɮ̪̪	ɮ̪̪	ɮ̪̪	ɮ̪̪̪
	<i>fric.</i>	<i>un.</i>	<i>asp.</i>	<i>un.</i>	<i>asp.</i>	<i>nas.</i>	<i>sv.</i>	<i>rh.</i>	<i>lat.</i>	<i>sh.</i>	<i>long.</i>		
		<i>plosive</i>					<i>gl.</i>	<i>liquid</i>		<i>mono.</i>		<i>diph.</i>	
							<i>approximant</i>			<i>vocalic</i>			
	<i>consonant</i>									<i>vowel</i>			

- *fric.* — fricative
- *un.* — unaspirated
- *asp.* — aspirated
- *nas.* — nasal
- *sv.* — semivowel
- *rh.* — rhotic
- *lat.* — lateral
- *sh.* — short
- *long.* — long
- *gl.* — glide
- *mono.* — monophthong
- *diph.* — diphthong

2.2 Sandhi

This section specifies all the rules of phonetic conjugation that exist in the language. Any rule that applied to an unaspirated consonant also applies to its aspirated equivalent, though this rule is not necessarily the same for voiced and voiceless counterparts of a sound.

2.2.1 Velars

Antecedent	Succedent	Result
ṭ	ḷ	ḷḷ
ḷ	ṭ	ḷḷ
ṭ	ḍ	ḍḍ
ḷ	ḍ	ḍḍ
ṭ □ ḷ	ḍ	ḍḍ
ṭ	ḷ	ḷḷ
ṭ	ṣ	ṣṣ
ḷ	ḷ, ṣ	ṣṣ
ṭ	ṣ	ṣṣ
ṭ	ṣ	ṣṣ
ḷ	ṣ, ṣ	ṣṣ
ḷ	ṣ	ṣṣ
ḷ	ṣ, ṣ	ṣṣ
ḷ	ṣ, ṣ	ṣṣ
ṭ	ṣ, ṣ	ṣṣ
ḷ	ṣ, ṣ	ṣṣ
ṭ	ṣ	ṣṣ
ḷ	ṣ	ṣṣ

2.2.2 Palatals

Antecedent	Succedent	Result
ḍ □ ḍ	ṭ □ ḷ	ṭṭ
ḍ	ḍ	ḍḍ
ḍ	ḍ	ḍḍ
ḍ	ḷ, ṣ	ḷḷ
ḍ	ḷ, ṣ	ḷḷ
ḍ	ḷ	ḷḷ
ḍ	ṣ	ṣṣ
ḍ	ḷ	ḷḷ
ḍ	ṣ	ṣṣ
ḍ, ḍ	ṣ	ṣṣ
ḍ, ḍ	ṣ	ṣṣ
ḍ	ḷ, ṣ	ḷḷ
ḍ	ḷ, ṣ	ḷḷ
ḍ, ḍ	ṣ	ṣṣ
ḍ	ṣ, ṣ, ṣ	ṣṣ
ḍ	ṣ, ṣ, ṣ	ṣṣ

2.2.3 Retroflexes

Antecedent	Succedent	Result
č	ʌ	ɽʌ
ɽ	ʈ	ɽʌ
č	ɖ	ɖɖ
č	ɛ	ɛɛ
ɽ	ɖ, ɛ	ɛɛ
č	ɽ	ɽɽ
ɽ	č	ɽɽ
č	ʌ	ʌʌ
č	ʂ	ʂʂ
ɽ	ʌ, ʂ	ʂʂ
č	ɔ	ɽɔ
ɽ	ʌ	ɽɔ
ɽ	ɪ	ɽɿ
čɔ ɽ	ʌ, ʌ	čɛ
č	ɟ, ɟ, ɽ	čɟ
ɽ	ɟ, ɟ, ɽ	ɽɟ

2.2.4 Dentals

Antecedent	Succedent	Result
ʌ	ʌ	ʂʌ
ʂ	ʈ	ʂʌ
ʌ	ɖ	ɖɖ
ʂ	ɖ	ɛɛ
ʌ, ʂ	ɛ	ɛɛ
ʌ	č	čč
ʂ	č	ɽɽ
ʌ, ʂ	ɽ	ɽɽ
ʌ	ʂ	ʂʂ
ʂ	ʌ	ʂʂ
ʌ	ɔ	ʂɔ
ʂ	ʌ	ʂɔ
ʌ, ʂ	ʌ	ɖɖ
ʌ, ʂ	ɛ	ɛɛ
ʌ	ɟ, ɟ, ɽ	ʌɟ
ʂ	ɟ, ɟ, ɽ	ʂɟ

2.2.5 Bilabial

Antecedent	Succedent	Result
ṽ	ṽ	ṽṽ
ṽ	ṽ	ṽṽ
ṽ	ṽ	ṽṽ
ṽ	ṽ	ṽṽ
ṽ	ṽ	ṽṽ
ṽ	ṽ	ṽṽ
ṽ	ṽ	ṽṽ
ṽ	ṽ	ṽṽ
ṽ	ṽ	ṽṽ
ṽ	ṽ	ṽṽ
ṽ	ṽ	ṽṽ
ṽ	ṽ	ṽṽ

2.2.6 Sibilants

The voiceless palato-alveolar sibilant $\tilde{\lambda}$ mutates into the voiceless retroflex sibilant $\tilde{\lambda}$ when initial to a velar, retroflex or a bilabial. Hetero-combinations of any of the sibilants (the above two and the voiceless coronal $\tilde{\lambda}$) results in a voiceless retroflex sibilant geminate.

2.2.7 Voiced Glottal Fricative/Aspirate

Whenever this sound ($\tilde{\lambda}$) is succedent to another velar, palatal, retroflex, dental or bilabial consonant, it leaves the initial alone and mutates into the aspirate (respecting the voice) of the initial.

2.2.8 Nasals

Antecedent	Succedent	Result
ṽ, ṽ, ṽ, ṽ	Velar	ṽ
ṽ, ṽ, ṽ, ṽ	Palatal	ṽ
ṽ, ṽ, ṽ, ṽ	Retroflex	ṽ
ṽ, ṽ, ṽ, ṽ	Dental	ṽ
ṽ, ṽ, ṽ, ṽ	Bilabial	ṽ
Velar	ṽ	ṽṽ
Palatal	ṽ	ṽṽ
Retroflex	ṽ	ṽṽ
Dental	ṽ	ṽṽ
Bilabial	ṽ	ṽṽ
Non-Labial Nasal 1	Non-Labial Nasal 2	Nasal 1 geminate

2.2.9 Vowels

- Any vowel encountering itself (even with different length) becomes a long vowel.
- The next set of rules combine vowel classes when the initial is a monophthong.
 - ṽ ṽ : — ṽ
 - ṽ ṽ ṽ — ṽ
 - ṽ ṽ ṽ or ṽ ṽ ṽ or ṽ ṽ ṽ — ṽ
 - ṽ ṽ ṽ or ṽ ṽ ṽ or ṽ ṽ ṽ — ṽ
 - ṽ ṽ ṽ + ṽ ṽ ṽ — ṽ
 - ṽ ṽ ṽ + ṽ ṽ ṽ — ṽ

- $\dot{\text{ɪ}} + \text{ɥ}, \text{ɥ} \rightarrow \text{ɔ}, \text{ɔ}$
- $\text{ɪ} + \text{ɥ}, \text{ɥ} \rightarrow \text{ɪɔ}, \text{ɪɔ}$
- $\dot{\text{ɪ}} + \text{ɮ}, \text{ɮ} \rightarrow \text{ɔ}, \text{ɔ}$
- $\text{ɪ} + \text{ɮ}, \text{ɮ} \rightarrow \text{ɪɔ}, \text{ɪɔ}$
- $\dot{\text{ɪ}} + \text{ɖ}, \Delta \rightarrow \text{ɔ}, \text{ɔ}$
- $\text{ɪ} + \text{ɖ}, \Delta \rightarrow \text{ɪɔ}, \text{ɪɔ}$
- $\dot{\text{ɪ}} + \text{ɭ}, \text{ɭ} \rightarrow \text{ɔ}, \text{ɔ}$
- $\text{ɪ} + \text{ɭ}, \text{ɭ} \rightarrow \text{ɪɔ}, \text{ɪɔ}$
- $\text{ɮ} + \text{ɥ}, \text{ɥ} \rightarrow \text{ɔ}, \text{ɔ}$
- $\text{ɮ} + \text{ɥ}, \text{ɥ} \rightarrow \text{ɮɔ}, \text{ɮɔ}$
- $\text{ɮ} + \dot{\text{ɪ}}, \text{ɪ} \rightarrow \text{ɔ}, \text{ɔ}$
- $\text{ɮ} + \dot{\text{ɪ}}, \text{ɪ} \rightarrow \text{ɮɔ}, \text{ɮɔ}$
- $\text{ɮ} + \text{ɖ}, \Delta \rightarrow \text{ɔ}, \text{ɔ}$
- $\text{ɮ} + \text{ɖ}, \Delta \rightarrow \text{ɮɔ}, \text{ɮɔ}$
- $\text{ɮ} + \text{ɭ}, \text{ɭ} \rightarrow \text{ɔ}, \text{ɔ}$
- $\text{ɮ} + \text{ɭ}, \text{ɭ} \rightarrow \text{ɮɔ}, \text{ɮɔ}$

- The next set of rules are for when the initial is a diphthong (either regular or monophthongised).
 - If the initial is a palatal, then it is left as it is and the final vowel is attached to $\dot{\text{ɔ}}$ (respecting length)
 - If the initial is a labial, then it is left as it is and the final vowel is attached to $\dot{\text{ɔ}}$ (respecting length)
 - In the above two cases, the resultant is long if and only if at least one of the two vowels is long.

Chapter 3

Grammar

3.1 Nominals

Given its Indo-Aryan origins, the grammar specifies an extended system of nine true nominal grammatical cases — nominative (default), accusative, instrumental, ablative, dative, inessive, adessive, subessive and vocative — and one pseudo-nominal adjectival genitive case. Unlike other Indo-Aryan languages, the vocative is treated as a true nominal case and not as a subsidiary nominative. The grammar allows for interesting recursive and paradoxical vocative usages. The system is fully nominative-accusative, unlike some ergative-absolutive Indo-Aryan descendants. There are three grammatical genders and three cardinal and ordinal numbers.

3.1.1 Nouns

Given in table 3.1 is a list of declensions taken by three words $\Gamma\mathcal{X}$, $\mathcal{J}\mathcal{K}$ and $\delta\mathcal{I}$, which are a boy's name, a girl's name, and a forest, in the masculine, the feminine and the neuter, respectively.

Case	Masculine	Feminine	Neuter
Nominative	$\Gamma\mathcal{X}$	$\mathcal{J}\mathcal{K}$	$\delta\mathcal{I}$
Accusative	$\Gamma\mathcal{X}$	$\mathcal{J}\mathcal{K}$	$\delta\mathcal{I}\mathcal{I}$
Instrumental	$\Gamma\mathcal{X}\mathcal{I}$	$\mathcal{J}\mathcal{K}\mathcal{I}$	$\delta\mathcal{I}\mathcal{I}\mathcal{I}$
Ablative	$\Gamma\mathcal{X}$	$\mathcal{J}\mathcal{K}\mathcal{I}$	$\delta\mathcal{I}\mathcal{I}\mathcal{I}$
Dative	$\Gamma\mathcal{X}\mathcal{I}$	$\mathcal{J}\mathcal{K}\mathcal{I}$	$\delta\mathcal{I}\mathcal{I}\mathcal{I}$
Genitive	$\Gamma\mathcal{X}\mathcal{I}$	$\mathcal{J}\mathcal{K}\mathcal{I}$	$\delta\mathcal{I}\mathcal{I}\mathcal{I}$
Inessive	$\Gamma\mathcal{X}$	$\mathcal{J}\mathcal{K}$	$\delta\mathcal{I}\mathcal{I}$
Adessive	$\Gamma\mathcal{X}\mathcal{I}$	$\mathcal{J}\mathcal{K}\mathcal{I}$	$\delta\mathcal{I}\mathcal{I}\mathcal{I}$
Subessive	$\Gamma\mathcal{X}\mathcal{I}$	$\mathcal{J}\mathcal{K}\mathcal{I}$	$\delta\mathcal{I}\mathcal{I}\mathcal{I}$
Vocative	$\Gamma\mathcal{X}$	$\mathcal{J}\mathcal{K}\mathcal{I}$	$\delta\mathcal{I}\mathcal{I}\mathcal{I}$

Table 3.1: Noun Declensions

3.1.2 Pronouns

Given in table 3.2 is a list of all the pronouns, of which three classes — personal, relative and interrogative.

Case	First	Second	Third	Relative	Interrogative
Nominative	ᖃ	ᖄ	ᖅ	ᖆ	ᖇ
Accusative	ᖃᖅ	ᖄ	ᖅ	ᖆ	ᖇ
Instrumental	ᖃ	ᖄ	ᖅ	ᖆ	ᖇ
Ablative	ᖃᖅ	ᖄᖅ	ᖅᖅ	ᖆᖅ	ᖇᖅ
Dative	ᖃᖅ	ᖄᖅ	ᖅᖅ	ᖆᖅ	ᖇᖅ
Genitive	ᖃᖅ	ᖄᖅ	ᖅᖅ	ᖆᖅ	ᖇᖅ
Inessive	ᖃ	ᖄ	ᖅ	ᖆ	ᖇ
Adessive	ᖃᖅ	ᖄᖅ	ᖅᖅ	ᖆᖅ	ᖇᖅ
Subessive	ᖃᖅ	ᖄᖅ	ᖅᖅ	ᖆᖅ	ᖇᖅ
Vocative	ᖃᖅ	ᖄᖅ	ᖅᖅ	ᖆᖅ	ᖇᖅ

Table 3.2: Pronouns

The three genders are intrinsic to a noun, but not to a pronoun and thus need to be suffixed separately, as specified in table 3.3.

Gender	Suffix
Masculine	(default)
Feminine	ᖃ
Neuter	ᖄ

Table 3.3: Grammatical Gender Suffixes

Both nouns and pronouns are inflected for three numbers in two forms as shown in table 3.4.

Type	Singular	Dual	Plural
Cardinal	(default)	ᖃ	ᖄ
Ordinal	ᖅ	ᖆ	ᖇ

Table 3.4: Number Suffixes

3.1.3 Declining Nominals

We have the following structure for declining a noun: *noun-root + case + number*, and for declining a pronoun: *pronoun-root + case + gender + number*. Note that inflecting for anything but grammatical case is done only once, when the topic of the sentence is introduced, or a new topic is injected. In future instances, unless deemed necessary, context is sufficient and the other inflections can be omitted. Here are a few examples to illustrate:

- ᖃᖅᖅ — in two/a pair of forests
- ᖃᖅᖅ — with whom (*feminine plural*)?

3.1.4 Numbers

Numbers by themselves are treated as pseudo-adjectives. They are inflected as: *number-root + gender + nature (cardinal/ordinal)*. The suffixes are identical. Table 3.5 lists out all the numbers in existence.

an aspect cannot stand by itself without a tense. The suffixes are listed in table 3.7. Compound moods are subsidiary and may be used as shown:

Mood	Suffix
Declarative	(default)
Conditional	ᄃ
Imperative	ᄅ
Interrogative	ᄇ
Abilitative	ᄉ
Exclamatory	ᄋ
Optative	ᄍ
Conditional–Interrogative	ᄏ
Conditional–Abilitative	ᄒ
Conditional–Optative	ᄔᄏ
Abilitative–Interrogative	ᄖ
Abilitative–Exclamatory	ᄘ
Optative–Interrogative	ᄚᄇ
Optative–Exclamatory	ᄚᄋ

Table 3.7: Mood Suffixes

Table 3.8 lists the suffixes used to mark person.

Person	Suffix
First	ᄛ
Second	ᄜ
Third	(default)

Table 3.8: Person Markers

Here’s an example to illustrate all the moving parts in action:

ᄃᄛᄋᄏᄉᄇᄉᄏᄏ — *they (feminine) couldn’t have played*

3.2.1 The Periphrastic Passive

So far, the verb structure used is purely in the active voice, which is usually the default tone. However, when necessary, the grammar specifies a passive voice, which can be formed in two ways. One (and the most common) way is by mutating the verb root itself into a passive form and then using the inflections specified by the active form, or (less commonly) by using the following structure:

instrumental subject + past form of the verb in agreement

Note that this method of forming the passive is only defined for the present tense, and does not exist for the past and the future tenses. For those, the verb root must be mutated. The past *form* means that the perfective aspect is carried over, as in, simple to simple and progressive to progressive. The verb in this new form must be inflected to agree with the subject in the instrumental form that comes with it. In this verb phrase, the order can be switched around, but the two components move together as a unit and cannot be separated. Such conveniences are only facilitated by mutating the verb root. When the subject is unspecified, use the neuter singular.

For example: ᄃᄛᄏᄏ ᄏᄏᄏᄏ — *it has been told*

3.3 Determiners & Conjunctions

3.3.1 Conjunctions

These are simple atoms that appear in between connecting clauses. Table 3.9 lists all the conjunctions.

and	ᄃ
or	ᄆ
but	ᄇ
enough/so/even	ᄃᄆ
than	ᄃ

Table 3.9: Conjunctions

3.3.2 Interrogatives & Relatives

For nuanced interrogative/relative words, they are formed by taking the inflected forms of the interrogative/relative pronouns and mutating their case inflections as given in table 3.10, but keeping the other inflections for number and gender the same. For most words, unless needed, the other inflections for number and gender can be dropped.

Word	Case	Intended Meaning
Why	Accusative	for what
How	Instrumental	by what
When	Dative	at what
Where	Locative (or its subcases)	in what

Table 3.10: Nuanced Interrogative/Relative Words

3.3.3 Proximity

When a third person personal pronoun is used with all of its inflections, it fails to specify the proximity of the specified with respect to the speaker and the listener. Table 3.11 lists the prefixes used to specify that.

Prefix	Proximity
(default)	unspecific/away from both the speaker and the listener
ᄃ	away from the speaker and close to the listener
ᄆ	close to the speaker, away from the listener
ᄃᄆ	close to both the speaker and the listener

Table 3.11: Proximity Markers

3.3.4 Clusivity

Because of three numbers, clusivity is far more intricate than usual. It is also marked by a prefix to a first person dual or plural personal pronoun. Table 3.12 lists all the clusivity markers.

Prefix	Meaning	Clusivity
(default)	speaker and direct listener; excluding passive listener	Dual Inclusive
ᄃ	speaker and passive listener; excluding direct listener	Dual Exclusive
ᄆ	including direct listener	Plural Inclusive
(default)	excluding direct listener	Plural Exclusive

Table 3.12: Clusivity Markers

Chapter 4

Lexicon

4.1 Conception & The Root System

While scouring the universe for various grammatical lexicon and word-building systems, I looked through various forms and practices in languages across the spectrum. Two systems, in particular, caught my eye, the Sanskrit–Dravidian root affixation system and the Semitic multi-consonantal template system. The Semitic system seemed very logical and straightforward. Although it was originally a similar affixing system just like the Indo–European equivalent, millenia of elision and evolution has turned it into this logical looking template system. A bit more research showed me that having an unnaturally obtrusive template system is not really good (obviously for the naturality) of the language, and so I settled for my native system.

Take the root $\pm$ in Sanskrit, which is associated with the action of *doing*. From it, by attaching various affixes (and millenia of sandhi and evolution), we get words like $\mathbf{d\bar{d}l}$ for *wheel*, $\mathbf{\bar{d}l\bar{x}}$ for *action*, $\mathbf{\bar{d}l\bar{i}}$ for *reason*, and $\mathbf{\bar{d}l\bar{a}}$ for *doër*. This language specifies a very similar, yet highly regular affixing system for taking atomic roots and converting them to grammatically sound and nuanced stems.

4.2 Derivation

Using the atomic bare roots as a base, there are many ways of affixing (subject to *sandhi*) it to deepen its nuance, mutate its meaning, or assign different grammatical functions to it. Every single nominal and verb in existence in the language is essentially an inflected affixed root stem. We will study the affix system by categorising affixes on the basis of grammatical function. Each section will begin with an affixing rule and a few examples after.

The affix system is divided into two classes: strong and weak, or class I and II, respectively. In a final grammatically valid sentence, there can only exist strong forms of any root. A strong form of a root is sort of the final form of the root, with all its nuances and embellishments injected, ready for either standing by itself or being inflected grammatically rather than semantically. A weak form of a root is an incomplete variant of the root, and needs to be semantically inflected further. It cannot stand by itself in a sentence and no grammatical affixes can be attached to a weak form directly. A class I affix produces a strong form, whereas a class II affix produces a weak form. In other words, a strong form is a stem, whereas a weak form is a root.

4.2.1 Infinitive (class I)

Rule: *root* + $\mathbf{\bar{a}l}$

- $\mathbf{\bar{d}l\bar{a}}$ — to do
- $\mathbf{\bar{d}l\bar{a}}$ — to see
- $\mathbf{\bar{d}l\bar{a}}$ — to die
- $\mathbf{\bar{d}l\bar{a}}$ — to purify
- $\mathbf{\bar{d}l\bar{a}}$ — to solidify/freeze
- $\mathbf{\bar{d}l\bar{a}}$ — to shine

answer ᐱᑦᑦᑦ
ant ᑦᑦ
anxiety ᐱᑦᑦ
any ᑦᑦᑦ
apart ᑦᑦᑦ
apex *top*
appear ᐱᑦᑦᑦ

apply ᐱᑦᑦᑦ
approach ᑦᑦᑦᑦ
arch ᐱᑦᑦᑦ
area ᐱᑦᑦᑦ
argue ᑦᑦᑦ
arm ᑦᑦ
army ᑦᑦ

around ᑦᑦ
arrange ᑦᑦ
arrive *come*
art ᑦᑦ
ash ᑦᑦ
ask ᑦᑦ
attack ᑦᑦᑦ

attempt ᑦᑦ
attract ᑦᑦ
authority *right*
autumn ᑦᑦᑦ
awake *sleep*
awe *wonder*
axis ᑦᑦᑦ

B

baby *immediate, life*
back ᑦᑦᑦ
bad *moral*
bag *hold*
bake *cook*
ball *round, play*
band ᑦᑦᑦ
bare ᑦᑦ
bark ᑦᑦᑦ
base ᑦᑦᑦ
bat ᑦᑦᑦ
bath ᑦᑦᑦ
be ᑦᑦᑦ
bear *fear, hair, animal*
beat ᑦᑦᑦ
beauty ᑦᑦᑦᑦ
bed *sleep*
bee *honey, create*
before *after*

begin ᑦᑦᑦᑦ
behind ᑦᑦᑦ
believe ᑦᑦᑦ
bell ᑦᑦᑦᑦ
belly ᑦᑦᑦ
below *above*
bend ᑦᑦᑦ
berry *fruit, size*
between *waist*
big *size*
bind ᑦᑦᑦᑦ
bird ᑦᑦ
birth *life*
bite ᑦᑦᑦ
bitter ᑦᑦ
blade ᑦᑦᑦ
blame ᑦᑦᑦ
blank ᑦᑦᑦ
blind *vision*

blood ᑦᑦ
blow ᑦᑦᑦ
board ᑦᑦᑦ
boat ᑦᑦᑦ
body ᑦᑦᑦ
boil ᑦᑦᑦᑦ
bone ᑦᑦᑦᑦ
book *knowledge, give*
border ᑦᑦᑦ
borrow *loan*
bottom *top*
bow ᑦᑦᑦᑦ
bowl *food, hold*
box *thing, hold*
boy *man, size*
brain ᑦᑦᑦᑦ
branch ᑦᑦ
brass ᑦᑦᑦᑦ

brave ᑦᑦ
bread ᑦᑦ
break ᑦᑦᑦ
breast *milk, create*
breath *man, air*
brick ᑦᑦᑦᑦᑦ
bridge ᑦᑦᑦ
bright *light*
bring *come*
broad *wide*
brother ᑦᑦᑦᑦ
brush *colour, tail*
build ᑦᑦᑦᑦ
burn ᑦᑦᑦ
burst *explode*
bury ᑦᑦᑦ
bush ᑦᑦ
busy ᑦᑦᑦᑦ
buy *receive, monetary*